

NEWPORT NEWS WILLIAMSBURG, VA, USA

WMO: 723086

Lat: 37.132N

Lon: 76.493W

Elev: 13

StdP: 101.17

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93741

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.0	-4.8	-17.4	0.8	-4.0	-14.6	1.1	-2.1	10.7	8.2	9.4	8.4	3.0	0	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	34.6	25.0	33.2	24.5	32.1	24.1	26.4	32.3	25.7	31.1	25.2	30.2	4.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.0	29.1	24.1	19.1	28.3	23.7	18.6	27.9	82.1	32.3	79.2	31.3	76.7	30.2	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	8.3	7.5	-10.9	37.3	3.2	2.0	-13.1	38.7	-15.0	39.9	-16.8	41.0	-19.1	42.4
			-11.9	28.0	2.8	1.1	-14.0	28.8	-15.6	29.4	-17.2	30.1	-19.3	30.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	4.6	6.0	9.6	15.2	19.7	24.3	26.5	25.6	22.5	16.6	10.7	6.9
	DBStd	8.89	5.96	5.30	5.26	4.89	4.27	3.49	2.76	2.59	3.24	4.38	4.77	5.27
	HDD10.0	582	188	135	74	9	0	0	0	0	0	3	47	126
	HDD18.3	1883	426	347	274	120	36	3	0	0	5	86	232	354
	CDD10.0	2682	21	22	62	164	302	431	512	484	374	209	69	31
	CDD18.3	939	1	1	3	26	79	183	254	226	129	34	3	1
	CDH23.3	8734	1	3	34	228	676	1774	2680	2150	952	219	14	4
	CDH26.7	3252	0	0	3	56	214	708	1149	809	267	44	1	0
Wind	WSAvg	3.6	3.7	3.8	4.1	4.1	3.7	3.5	3.4	3.2	3.3	3.3	3.3	3.4
Precipitation	PrecAvg	1191	94	91	106	74	102	106	124	122	120	80	78	90
	PrecMax	1727	221	150	278	213	220	266	268	249	394	306	169	194
	PrecMin	718	18	6	27	12	11	14	31	15	14	9	7	11
	PrecStd	253	38	35	52	43	51	63	60	63	89	57	46	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	23.6	27.1	30.8	33.0	36.2	36.8	35.4	33.4	30.2	25.1	23.6
		MCWB	17.3	17.2	17.6	20.0	23.1	25.0	25.6	25.6	23.7	21.9	18.8	18.1
	2%	DB	19.7	20.3	23.8	28.3	31.1	33.9	34.9	33.7	31.4	27.8	22.9	20.8
		MCWB	16.0	14.4	16.0	18.9	22.4	24.5	25.4	24.9	23.3	21.1	17.6	17.1
	5%	DB	17.3	17.7	21.3	26.2	29.1	32.3	33.4	32.3	29.6	25.9	21.1	18.0
		MCWB	13.8	13.4	14.7	18.0	21.4	23.9	25.0	24.4	22.8	20.2	17.0	15.2
	10%	DB	14.1	15.1	18.8	23.6	27.3	30.8	32.1	31.0	27.9	23.9	18.8	15.8
		MCWB	11.0	10.7	13.8	17.1	20.5	23.4	24.6	24.0	22.2	19.3	15.4	12.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	18.0	19.3	21.9	24.7	26.3	27.4	27.4	25.7	24.1	20.9	19.8
		MCDB	21.5	21.7	24.1	27.6	30.3	32.6	33.4	33.2	30.2	26.4	22.9	22.5
	2%	WB	16.5	16.3	17.7	20.5	23.5	25.5	26.7	26.3	24.9	22.8	19.1	17.5
		MCDB	19.2	19.4	21.5	25.9	28.8	31.7	32.7	31.5	28.8	26.0	21.7	19.9
	5%	WB	14.2	14.0	16.3	19.3	22.5	24.8	25.9	25.6	24.3	21.6	17.6	15.6
		MCDB	16.5	17.1	20.5	24.3	27.4	30.5	31.6	30.4	27.7	24.8	20.0	17.8
	10%	WB	11.4	11.2	14.4	18.0	21.5	24.2	25.3	24.9	23.5	20.1	15.8	13.2
		MCDB	13.8	14.6	18.2	22.5	26.1	29.1	30.5	29.3	26.6	23.2	18.5	15.6
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.5	10.9	11.5	10.6	10.0	9.7	9.4	9.5	10.6	10.8	9.8
		MCDBR	13.0	14.1	14.2	13.6	12.2	11.2	10.9	10.5	10.8	11.8	12.5	11.8
		MCWBR	9.9	9.9	8.6	6.6	5.8	4.5	4.1	4.1	4.7	6.4	8.0	9.1
	5% WB	MCDBR	11.9	13.2	12.4	12.1	10.8	10.0	9.9	9.1	8.5	9.2	10.4	11.0
		MCWBR	10.2	10.6	9.2	6.8	5.6	4.4	4.4	4.1	4.3	5.5	8.0	9.4
Clear-Sky Solar Irradiance	taub	0.306	0.318	0.345	0.392	0.434	0.485	0.514	0.509	0.409	0.353	0.322	0.310	
	taud	2.541	2.502	2.456	2.337	2.249	2.156	2.090	2.091	2.361	2.513	2.551	2.562	
	Ebn at Noon	890	920	920	890	853	807	779	773	842	867	863	861	
	Edh at Noon	77	90	104	124	138	151	160	157	113	89	76	71	
All-Sky Solar Radiation	RadAvg	2.34	3.09	4.12	5.22	5.89	6.35	6.05	5.46	4.43	3.51	2.64	2.08	
	RadStd	0.19	0.34	0.33	0.38	0.53	0.42	0.30	0.29	0.38	0.41	0.22	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+141	+99
Regional (68 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+161	+104

Nomenclature: See separate page